MILITARY SPECFICATION

URANIUM TETRAFLUORIDE (UF_4), DEPLETED

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers depleted uranium tetrafluoride (UF_4) which is intended to be reduced to depleted uranium metal for ordnance use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

FSC 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS

TITLE 10 CFR - Safe Handling Procedures of Uranium
TITLE 49 CFR - Shipment of Uranium

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B-214 - Sieve Analysis of Granular Metal Powders, Test For
ASTM-B-329 - Apparent Density of the Refractory Metals

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Composition. The depleted uranium tetrafluoride shall be a minimum of 98.7 percent UF_4 and shall comply with the maximum impurity contents specified in Table I.

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TABLE I. UF_4 COMPOSITION LIMITS

ELEMENT	MAXIMUM PERMISSIBLE AMOUNT
Iron (Fe)	25 PPM
Nickel (Ni)	25 PPM
Copper (Cu)	20 PPM
Silicon (Si)	40 PPM
Aluminum (Al)	30 PPM
Manganese (Mn)	30 PPM
Carbon (C)	15 PPM
Chromium (Cr)	15 PPM
Ammonium Oxalate Insolubles (AOI)	0.3 Percent
UO_2F_2	1.0 Percent
UO_2F_2 + AOI	1.0 Percent
U_{235}	0.3 Percent

3.1.1 Properties. The depleted uranium tetrafluoride shall comply with the property requirements specified in Table II.

TABLE II. UF_4 PROPERTY REQUIREMENTS

PROPERTY	REQUIREMENTS	APPLICABLE TEST METHOD
Bulk Density	175 ± 35 pounds per cubic foot	4.6.2
Particle Size	100% shall pass through an 8 mesh screen and 90% through a 40 mesh screen	4.6.3
Moisture Content	0.03 percent, max by weight	4.6.4

3.2 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Process control documents (PCD). The contractor shall prepare a process control document which shall include descriptions of all production processes, procedures and equipment used to manufacture and inspect the production material as stipulated in the contract. The descriptions shall identify, as a

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minimum, the sequence of operations and schematic line layouts, basic fabrication processes, fabrication equipment, controls associated with the fabrication processes (i.e., times, temperature, pressures, etc.), associated processes and controls, inspection procedures, and inspection equipment. The PCD shall be submitted in accordance with DI-P-1604. All initial and revised PCD's shall be submitted to DRSMC-QAT-A(D) for review and disposition.

3.4 Identification. Each drum of UF_4 shall be marked with an identification number which indicates the UF_4 lot number and the sequential order in which the drums were filled. The contractor shall submit his identification numbering system to DRSMC-QAT-A(D) for approval prior to the start of production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article shall consist of a single lot of UF_4 (see 4.4.1) weighing between 25,000 - 30,000 lbs. This lot shall be produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The first article sample (see 4.5.1.1) will be subjected by the Government to any or all of the examinations or tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of UF_4 produced by one manufacturer, in accordance with the same specification or specification revision, under one continuous set of operating conditions, utilizing UF_6 from a single manufacturer, and limited to a maximum weight of 300,000 lbs. A new lot shall be formed whenever the continuous production cycle is interrupted. Each lot of UF_4 shall be identified as to the date and time of the production cycle start and finish. In addition, the identity of all UF_6 cylinders contributing to each lot of UF_4 shall be documented.

4.4.1.1 Regrouped lot. The formation of a regrouped lot shall be permitted in the event that a lot, or portion of a lot, fails to meet the requirements (Section 3.0) of this specification. A regrouped lot shall be formed by blending a non-conforming quantity of UF_4 with another quantity of UF_4 of known characteristics. The resultant UF_4 shall be identified as a regrouped lot and shall be inspected for all requirements of 3.0.

4.5 Testing.

PRECAUTION WARNING

This specification covers the sampling and testing of radioactive material. Accordingly, all applicable safety rules, regulations and procedures must be followed in handling and processing these materials. Regulations defining safe handling procedures are contained in the U.S. NRC Regulatory Guides 8.9, 8.10, 8.11, 8.13, 8.15 and Title 10 Code of Federal Regulations, Parts 19, 20, 40 and other parts as applicable.

4.5.1 Sampling.

4.5.1.1 First article inspection. Representative samples shall be randomly selected from 50% of the drums of UF_4 from the first article lot and shall be tested in accordance with 4.6.1, 4.6.2, 4.6.3, and 4.6.4.

4.5.1.2 Quality conformance inspection. A representative sample shall be selected from each consecutively filled drum of UF_4. The sample from the first drum filled at the beginning of each 24 hour cycle of continuous production and the last drum filled whenever the production cycle is interrupted shall be tested in accordance with 4.6.1, 4.6.2, 4.6.3, and 4.6.4. Failure of any sample to meet the applicable requirements shall be cause

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for 100% screening of all subsequently filled drums until 4 consecutive samples are tested successfully. The 100% screening shall only be performed with respect to those requirements which were not met initially. In addition, 100% screening shall also be performed in reverse production sequence until a sample meets the applicable requirements or until a previously accepted drum of UF_4 is reached. Testing at the beginning of each 24 hour cycle shall resume after 4 consecutive samples are found to meet the applicable requirements.

Rejected drums of UF_4 may be blended into a regrouped lot where possible in accordance with 4.4.1.1 and 4.5.1.2.1.

The contractor may dispose of any untested samples after disposition has been finalized for a given lot of UF_4.

4.5.1.2.1 Regrouped lots. Representative samples from each regrouped lot shall be randomly selected from 20% of the drums in the lot and tested in accordance with 4.6.1, 4.6.2, 4.6.3, and 4.6.4. Failure of any sample to meet the applicable requirements shall be cause for rejection of the regrouped lot.

4.5.2 Inspection equipment. The contractor shall use commercial inspection equipment for all of the tests and examinations specified in 4.6. The contractor shall have the required equipment available, follow proper calibration methods and perform the testing in accordance with accepted operating procedures. All inspection equipment designs shall be submitted for approval by the contractor in accordance with the terms of the contract. Submission shall be in accordance with DI-R-1714 with Addendum I.

4.6 Methods of inspection.

4.6.1 Chemical composition. Chemical analysis of each required sample shall be performed to determine compliance with the requirements of paragraph 3.1 and Table I. Samples shall be analyzed utilizing standard laboratory procedures and equipment approved by DRSMC-QAT-I(D).

4.6.2 Bulk density. Each required sample shall be analyzed for compliance with the bulk density requirement of Table II. Bulk density shall be determined in accordance with the methods described in ASTM B-329, except that the required 16 mesh screen shall not be used.

4.6.3 Particle size. Each required sample shall be analyzed for compliance with the particle size requirements of Table II. Particle size shall be determined by analysis of samples in accordance with the methods described in ASTM B-214.

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4.6.4 Moisture content. Each required sample shall be analyzed for compliance with the moisture content requirement of Table II. Moisture content shall be determined by weighing a sample between 20 - 100 grams prior to and after heating it to 230° ± 9°F for a sufficient duration of time to arrive at a stable condition. The heating cycle shall be repeated, as required, until the difference in weight between successive weighings is no more than .005% of the sample weight. Percent moisture content shall be calculated as follows:

$$P = 100\ (W-D)/W$$

where:
P = moisture content of sample, percent,
W = weight of original sample, g. and
D = final weight of dried sample, g.

5. PACKAGING

5.1 Packaging. UF_4 shall be packaged in metal drums meeting the requirements of DOT 37P, DOT 17C, DOT 17E or DOT 17H. If DOT 17C, 17E, or 17H drums are used, they shall be polyethylene lined in the same manner as that required for DOT 37P drums. The contractor shall also comply with all applicable Nuclear Regulatory Commission, Department of Energy, Department of Transportation, State and local regulations and licenses.

5.2 Marking. Marking shall be in accordance with MIL-STD-129 and Title 49 of the Code of Federal Regulations.

6. NOTES

6.1 Intended use. The UF_4 ordered to this specification is intended for use in reduction to depleted uranium metal of high purity which will be subsequently melted to produce a uranium alloy of specified purity for ordnance use.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures or equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Center (ARDC), ATTN: DRSMC-QAT-I(D), Dover, NJ 07801.

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6.4 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

PARAGRAPH	DATA REQUIREMENTS	APPLICABLE DID
3.3	Manufacturing Methods Report	DI-P-1604
4.5.2	Quality Engineering Acceptance Inspection Equipment Description Documentation	DI-R-1714

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 6810-AB37)

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